

Building blocks

Your body is made of trillions of cells. These are living building blocks that are really tiny. The cells shown here have all been photographed with the help of a powerful microscope to make them visible. Cells are constantly dividing to produce new cells that enable your body to grow and to repair itself.

Unlike other body cells, a stem cell does not have a distinctive shape because it does not have a specialized role.

THE CELL FACTORY

All cells inside your body are produced by stem cells. These stem cells are unique because they can change into the many different types of cells that make and shape your body, including skin, muscle, and blood cells. Cell production is a job that starts in the earliest days of life.

FAST FACTS

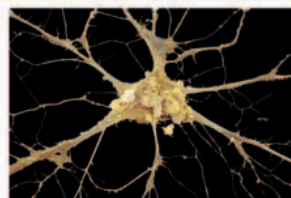
- The body's biggest cells—egg cells or ova—are 0.004 in (0.1 mm) across and just visible without a microscope.
- The smallest body cells are just 0.00016 in (0.004 mm) long and are found in the cerebellum (part of the brain).
- Placed in a row, 40 average-sized cells would stretch across a period.

TYPES OF CELLS

There are more than 200 types of cell in the body, six of which are shown here. Each has its own specific shape and size, related to the job it does. Nerve cells, for example, are long and thin, ideal for carrying electrical signals. Fat cells, crammed with fat droplets, provide fuel for the body and help insulate and keep you warm.



▲ **EPITHELIAL CELLS**
Protective epithelial cells—these are from an air passage—line flat surfaces and organs.



▲ **NERVE CELL**
Wirelike nerve cells (neurons) carry electrical signals and form the nervous system.



▲ **FAT CELL**
Also called adipocytes, these rounded cells store fat, one of the body's energy sources.